

SOLID STATE PHYSICS SEMINAR

INTRODUCTION:

In solid-state physics, we often begin by assuming a perfect crystal, where atoms are arranged in a completely periodic and ordered structure. However, in reality, no crystal is perfectly ordered. Due to thermal motion and energy fluctuations, all crystals contain imperfections known as defects.

Defects are not rare — they are naturally present and essential for determining material properties.

“Why should we even care about defects?”

Because it cause, many important properties like conductivity and diffusion arise because of these imperfections.

Types of defects:

Defects in crystals are generally classified based on their dimensionality. They can be categorized as point defects, line defects, and surface defects. Among these, point defects are the simplest and most fundamental, as they involve irregularities at a single lattice point.

The two most important types of point defects in ionic crystals are Schottky defects and Frenkel defects.

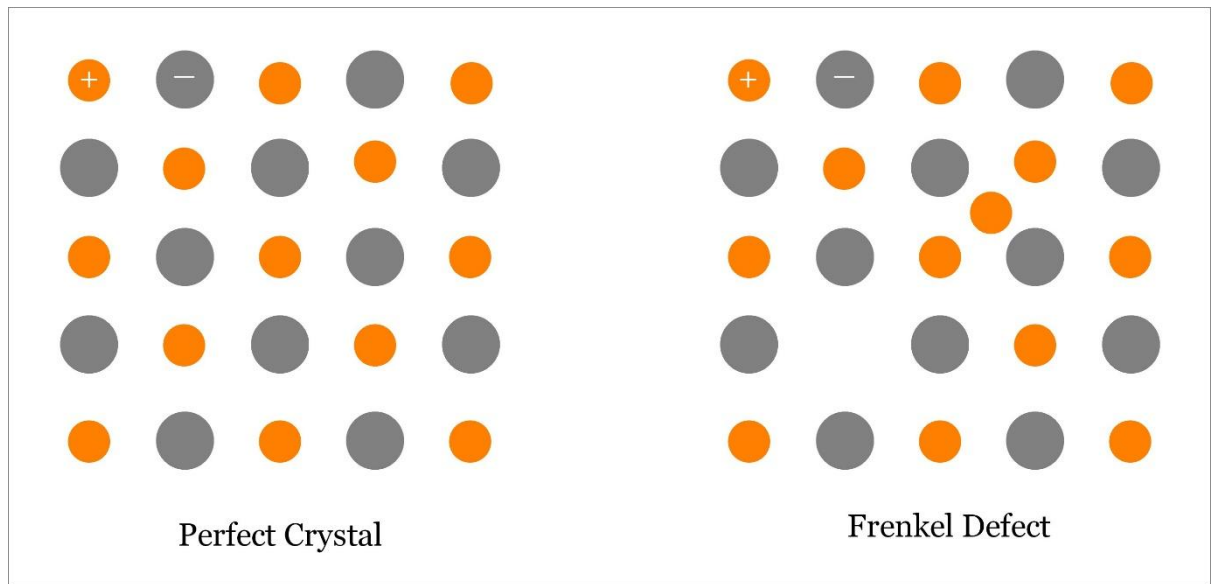
classification

- Point defects → (vacancy, interstitial, Frenkel, Schottky)
- Line defects → dislocations
- Surface defects → grain boundaries

While Schottky defects involve missing ions from the lattice, Frenkel defects involve displacement of ions within the crystal.

FRANKEL DEFECT:

A Frenkel defect is a type of point defect in which an atom, usually a smaller cation, leaves its normal lattice position and occupies an interstitial site. This results in the formation of a vacancy–interstitial pairs



Here, the atom moves from its original position, creating a vacancy. It then occupies an interstitial site, which is a space between the regular lattice positions.

KEY POINT:

There are some important characteristics of Frenkel defects:

- There is no change in density
- Electrical neutrality is maintained
- It forms a vacancy–interstitial pair

This type of defect is common in ionic crystals where the cation is small and can easily fit into interstitial spaces, such as silver halides like AgCl.

why do Frenkel defects form at all, if they disturb the crystal structure?

The formation of Frenkel defects is governed by a balance between energy and entropy.

To create a Frenkel defect, energy is required to move an atom from its lattice position to an interstitial site. This energy is called the defect formation energy.

However, when a defect is created, the number of possible atomic arrangements in the crystal increases. This increases the entropy of the system.

energy tries to minimize defects, while entropy tries to increase them.

At equilibrium, the system minimizes its free energy, which is given by $F = U - TS$.

To calculate how many Frenkel defects are present in a crystal at equilibrium, we use a statistical approach based on the number of possible atomic arrangements.

Let N be the total number of atoms, N_i be the number of interstitial sites, and n be the number of Frenkel defects.

$$W = \frac{N!}{n!(N-n)!} * \frac{N_i!}{n!(N_i-n)!} \quad (1)$$

Where,

- W Number of possible arrangements
- $N!$ total number of interstitial sites
- N total number of atoms in a crystal
- n number of Frenkel defects

The corresponding increase in entropy from Boltzmann relation

$$S = k_B \ln W_i \quad (2)$$

$$S = k_B \log \left[\frac{N!}{n!(N-n)!} * \frac{N_i!}{n!(N_i-n)!} \right]$$

$$= k_B \left[\log \frac{N!}{n!(N-n)!} * \log \frac{N_i!}{n!(N_i-n)!} \right]$$

As per the logarithmic

$$\log \left(\frac{A}{B} \right) = \log A - \log b$$

$$\log(AB) = \log A + \log b$$

$$S = k_B [\log N! - \log (N-n)! - \log n! - \log N_i! - \log n! - \log (N_i-n)!]$$

Using Stirling Approximation

$$\log N! = N \log N - N$$

$$S = k_B [N \log N - N - (N - n) \log (N - n) + (N - n) - n \log n + n + N_i \log N_i - N_i - n \log n + n - (N_i - n) \log (N_i - n) + (N_i - n)]$$

$$S = k_B [N \log N - (N - n) \log (N - n) - n \log n - N_i \log N_i - n \log n - (N_i - n) \log (N_i - n) - N + (N - n) + n - N_i + n + (N_i - n)]$$

$$S = k_B [N \log N - (N - n) \log (N - n) - n \log n - N_i \log N_i - n \log n - (N_i - n) \log (N_i - n) - N + N - n + n - N_i + n + N_i - n]$$

$$S = k_B [N \log N - (N - n) \log (N - n) - 2n \log n - N_i \log N_i - (N_i - n) \log (N_i - n)]$$

The change in Helmholtz free energy is

$$F = U - TS =$$

$$F = nE_i - k_B T [N \log N - (N - n) \log (N - n) - 2n \log n - N_i \log N_i - (N_i - n) \log (N_i - n)]$$

$$\frac{\partial F}{\partial n} = E_i - k_B T [N \log N - (N - n) \log (N - n) - 2n \log n - N_i \log N_i - (N_i - n) \log (N_i - n)] = 0$$

$$E_i = k_B T \left[+ \left(\frac{N - n}{N - n} \right) + \log (N - n) - 2 \log n - 2 + \left(\frac{N_i - n}{N_i - n} \right) + \log (N_i - n) \right]$$

$$E_i = k_B T [+1 + \log (N - n) - \log n^2 - 2 + 1 + \log (N_i - n)]$$

$$E_i = k_B T [\log (N - n) - \log n^2 + \log (N_i - n)]$$

$$E_i = k_B T \log \left[\frac{(N - n)(N_i - n)}{n^2} \right]$$

For $n \ll N, N_i$

$$E_i = k_B T \log \left[\frac{NN_i}{n^2} \right]$$

$$\frac{n^2}{NN_i} = \exp \left(-\frac{E_i}{2k_B T} \right)$$

$$n = \sqrt{NN_i} \exp \left(-\frac{E_i}{2k_B T} \right)$$

By minimizing the Helmholtz free energy with respect to the number of defects, we obtain the equilibrium concentration of Frenkel defects. The result shows that defect concentration depends exponentially on temperature, indicating that defect formation is a thermally activated process.